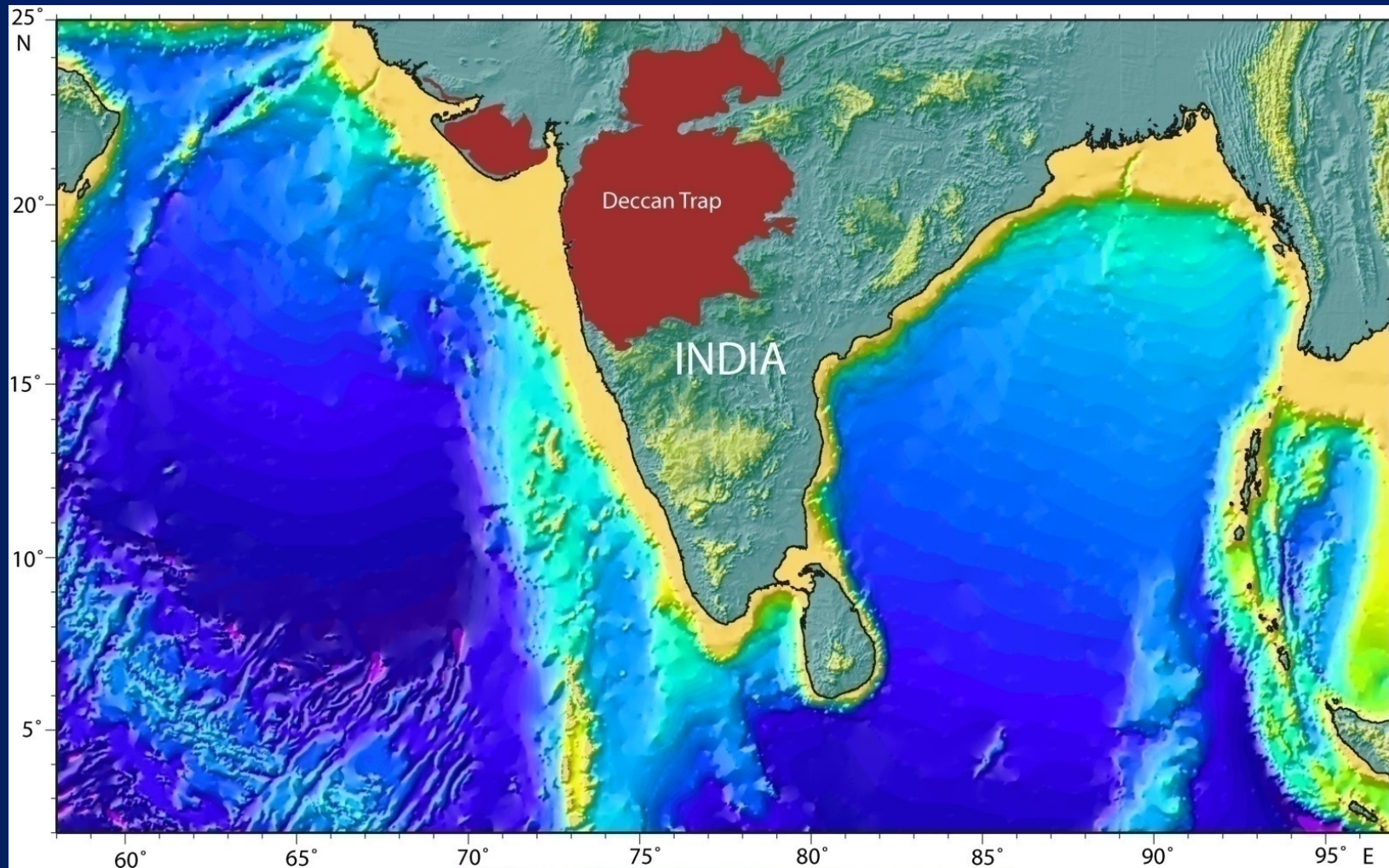


El Nino, La Nina and Southern Oscillation



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El Nino, La Nina and Southern Oscillation

El Niño /La Niña and Southern Oscillation together is known as ENSO.

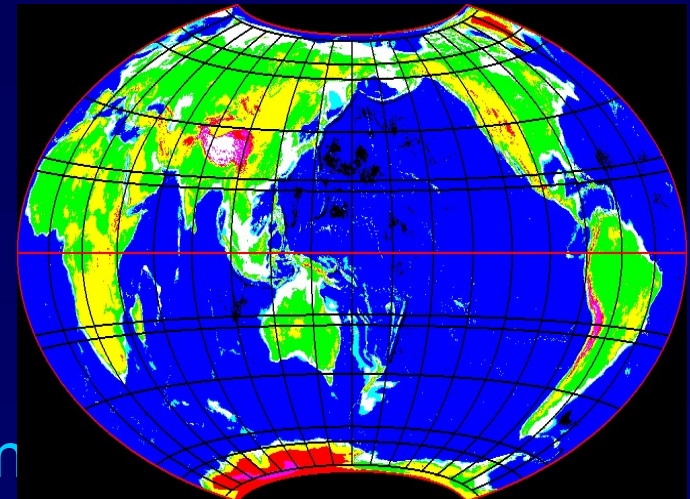
ENSO is a climate phenomenon that occurs in the Pacific ocean. To understand ENSO we need to understand the phenomenon of El Niño / La Niña & Southern Oscillation.

El Niño/La Niña is an oceanic phenomenon, while Southern Oscillation is an atmospheric phenomenon.

El Niño refers to warmer than normal condition of the Eastern Pacific.

La Niña refers to colder than normal condition of the Eastern Pacific.

El Niño and La Niña are results of fluctuation of sea-surface temperature in equatorial western and eastern Pacific.



ENSO Cycle



1. Neutral Cycle / Normal Cycle

2. La Nina

3. El Nina

Key Concepts

$T \uparrow \rightarrow P \downarrow$

$T \uparrow \rightarrow E \uparrow$

This makes air to ascend, as a result the air expand and cool.

It gives rise to condensation of water vapour in air mass and cloud formation.

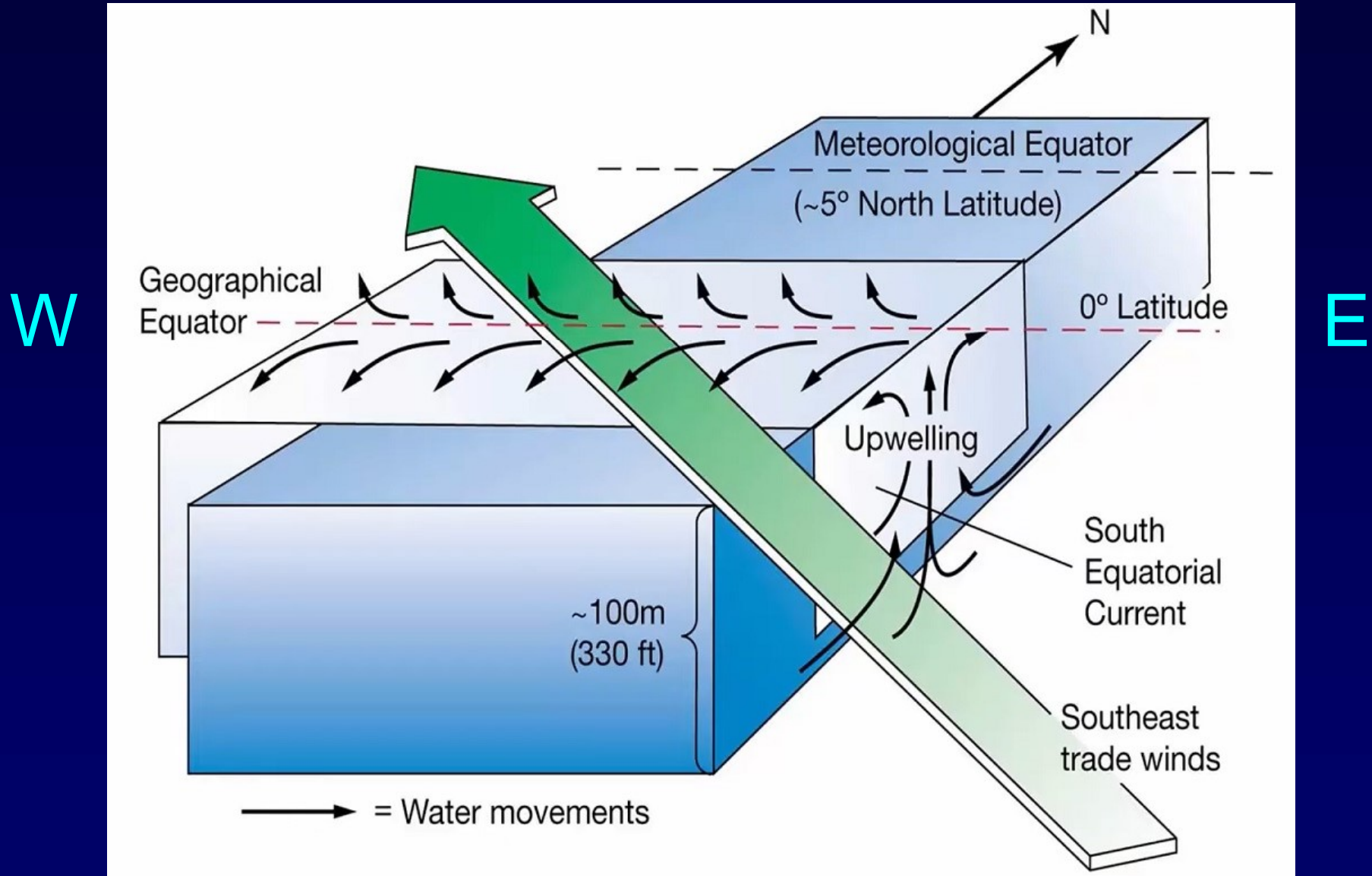
Finally, it rains.

1. Normal conditions in the Equatorial Pacific

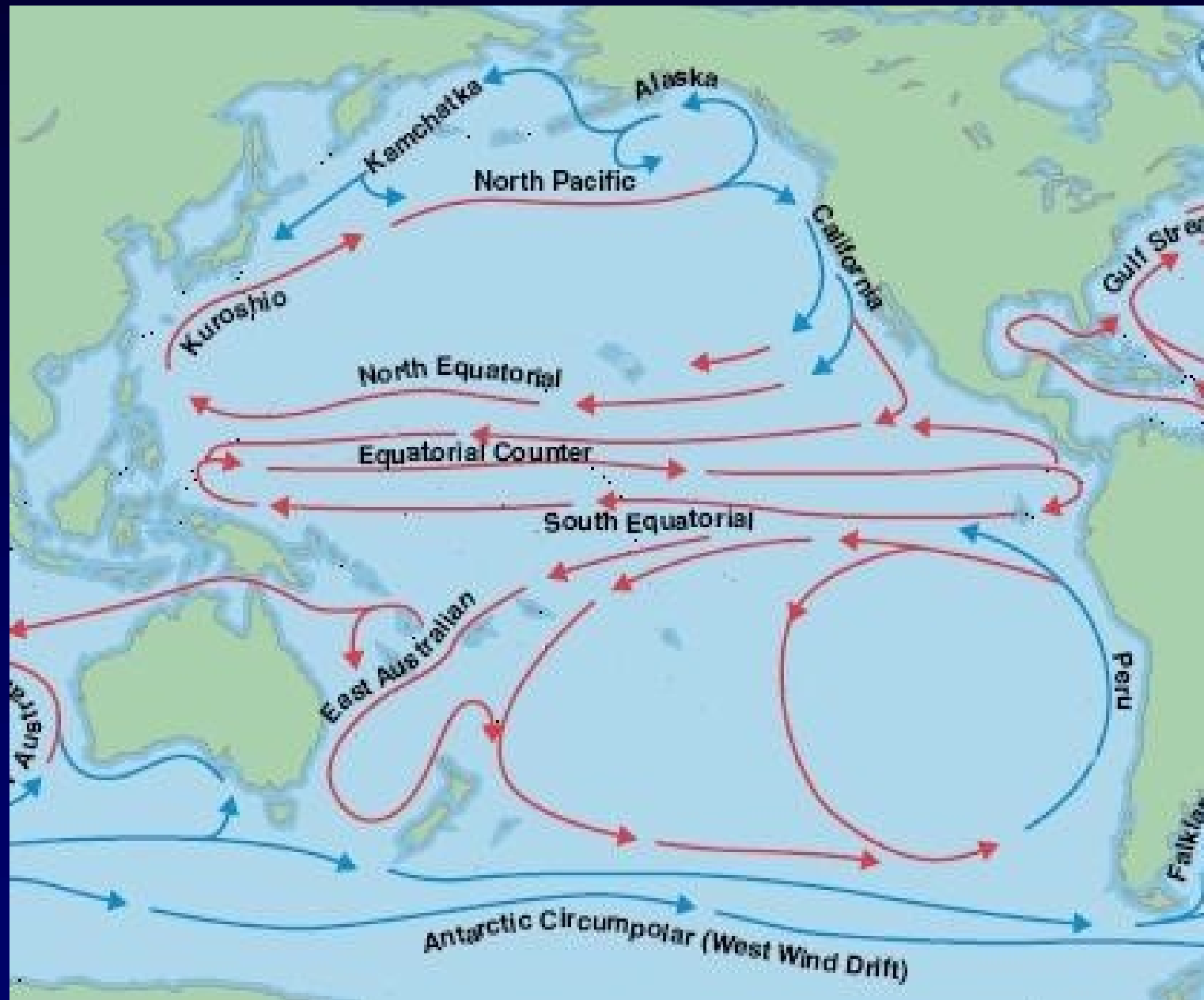
During normal years the Western Equatorial Pacific is warmer and the Eastern Equatorial Pacific is colder. This could be understood in the context of prevailing winds and Ocean circulation.

- Trade Winds blow from east to west. The N. Easterly and S. Easterly trade winds drive the NEC and SEC in the Equatorial Pacific. These currents transports and piles up warm waters at the Western Equatorial Pacific making this region much warmer.
- In the Eastern Equatorial Pacific, the south Easterly trade winds drive upwelling off Peru coast. This cold up welled water is transported offshore making Eastern Equatorial Pacific much colder.
- Movement of water to the west causes deep water to up-well up just offshore Ecuador and Peru.
- Deep water carries nutrients for phytoplankton. Thriving phytoplankton attract zooplankton, fish and fisherman.

1. Normal conditions in the Equatorial Pacific.....Contd.

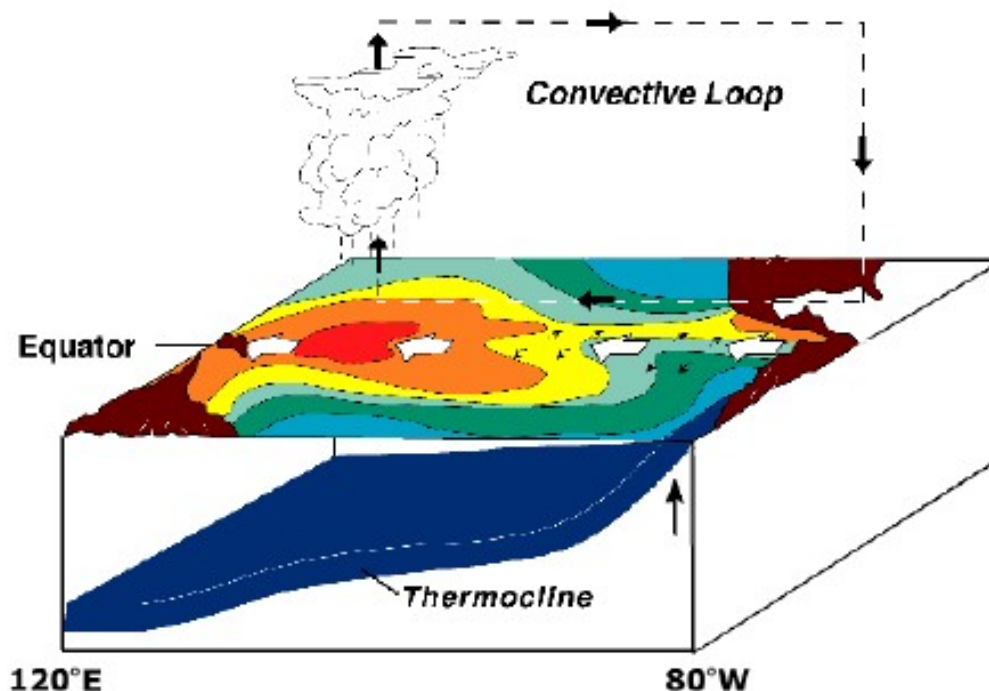
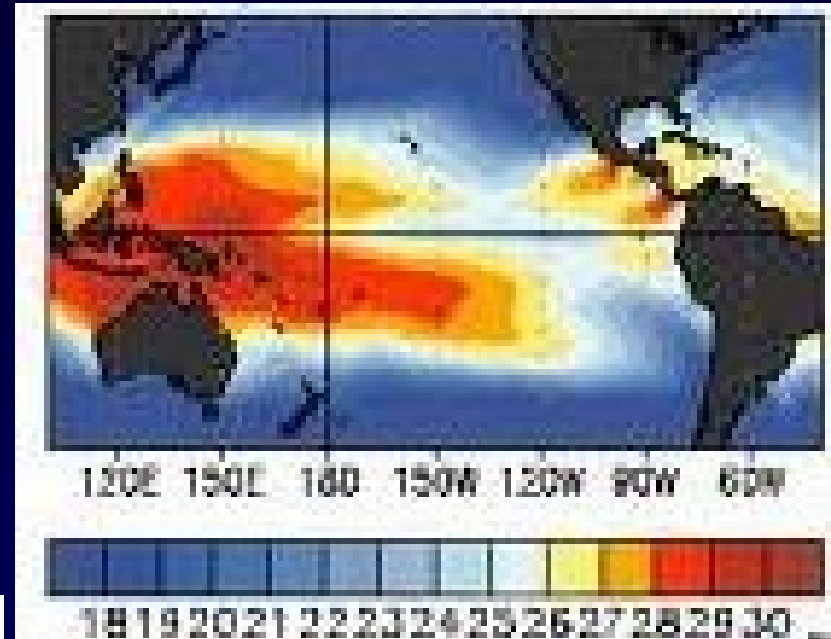


Equatorial Current in the Pacific Ocean



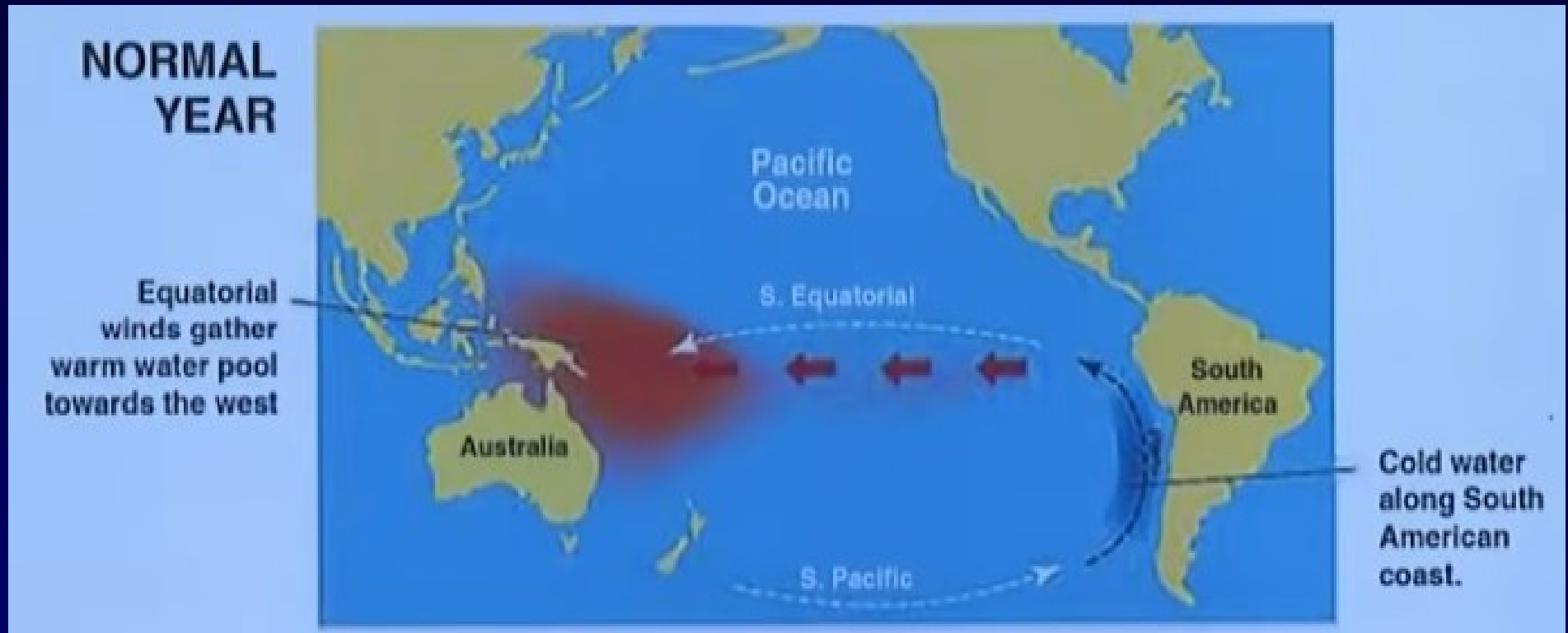
1. Normal conditions in the Equatorial Pacific

Thus in a normal year, the Eastern Equatorial Pacific is colder while the Western Equatorial Pacific is warmer.

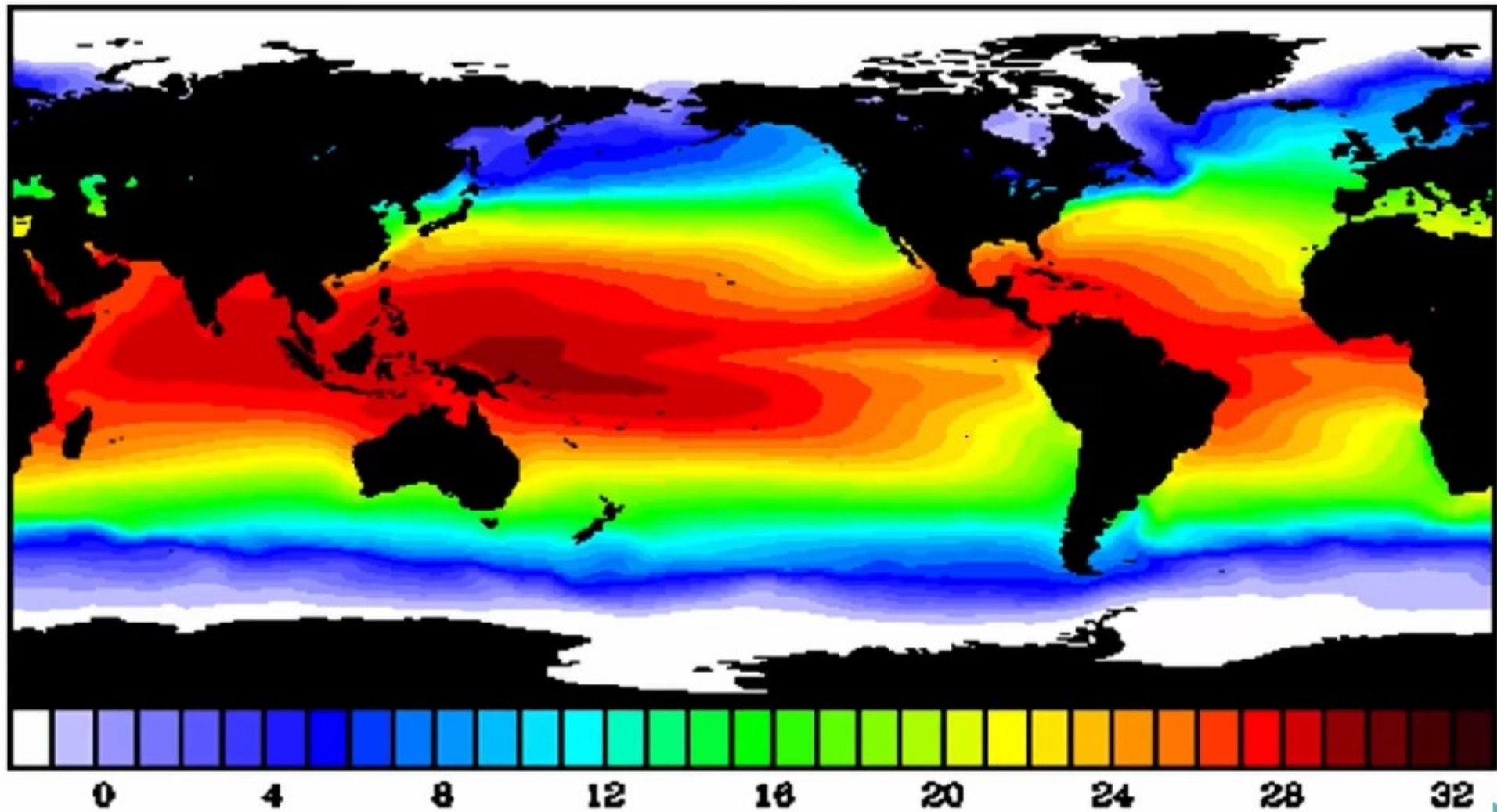


Equatorial winds gather warm water pool toward west. Cold water upwells along South American coast.

1. Normal conditions in the Equatorial Pacific



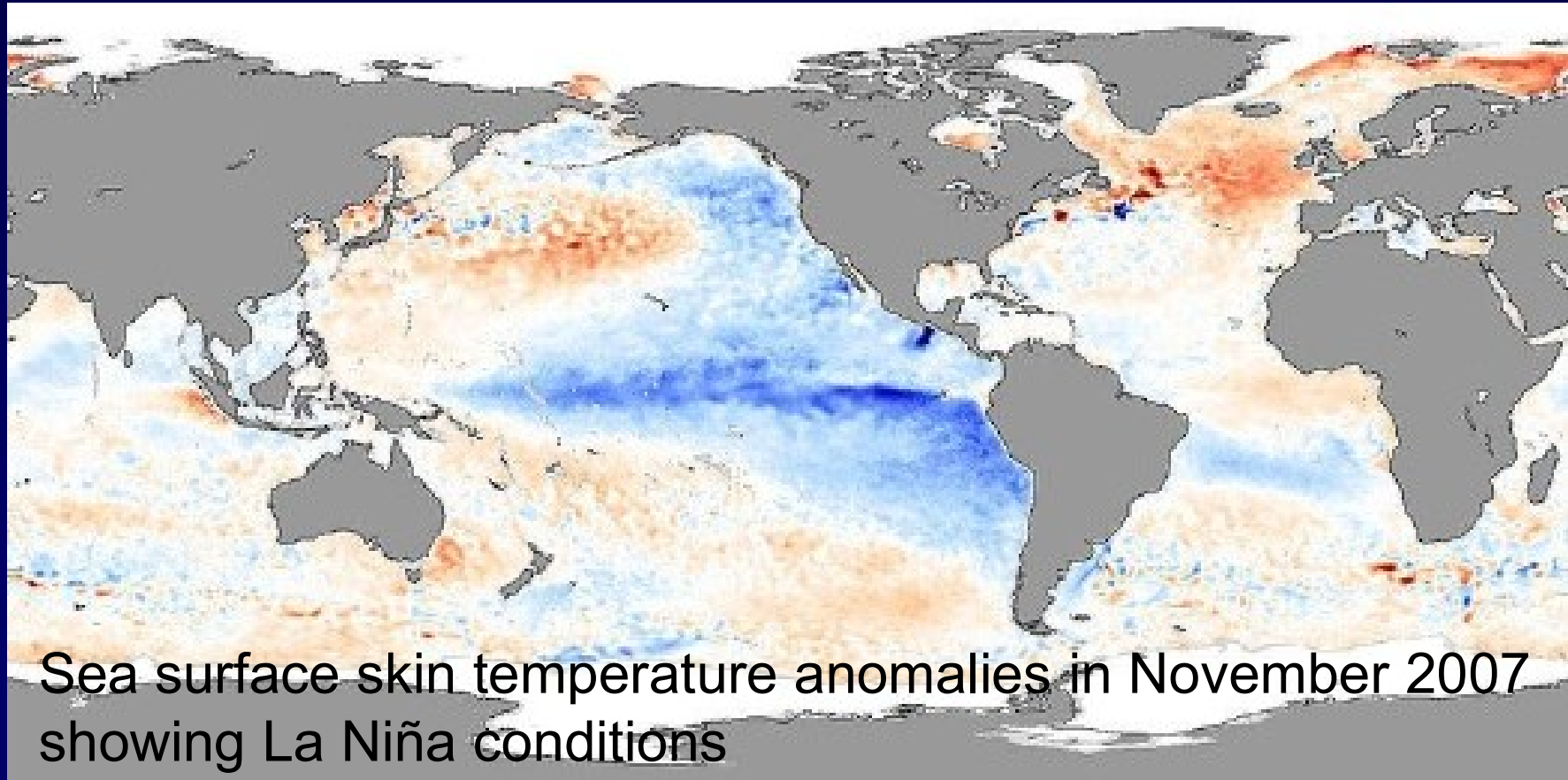
SST in Equatorial Pacific



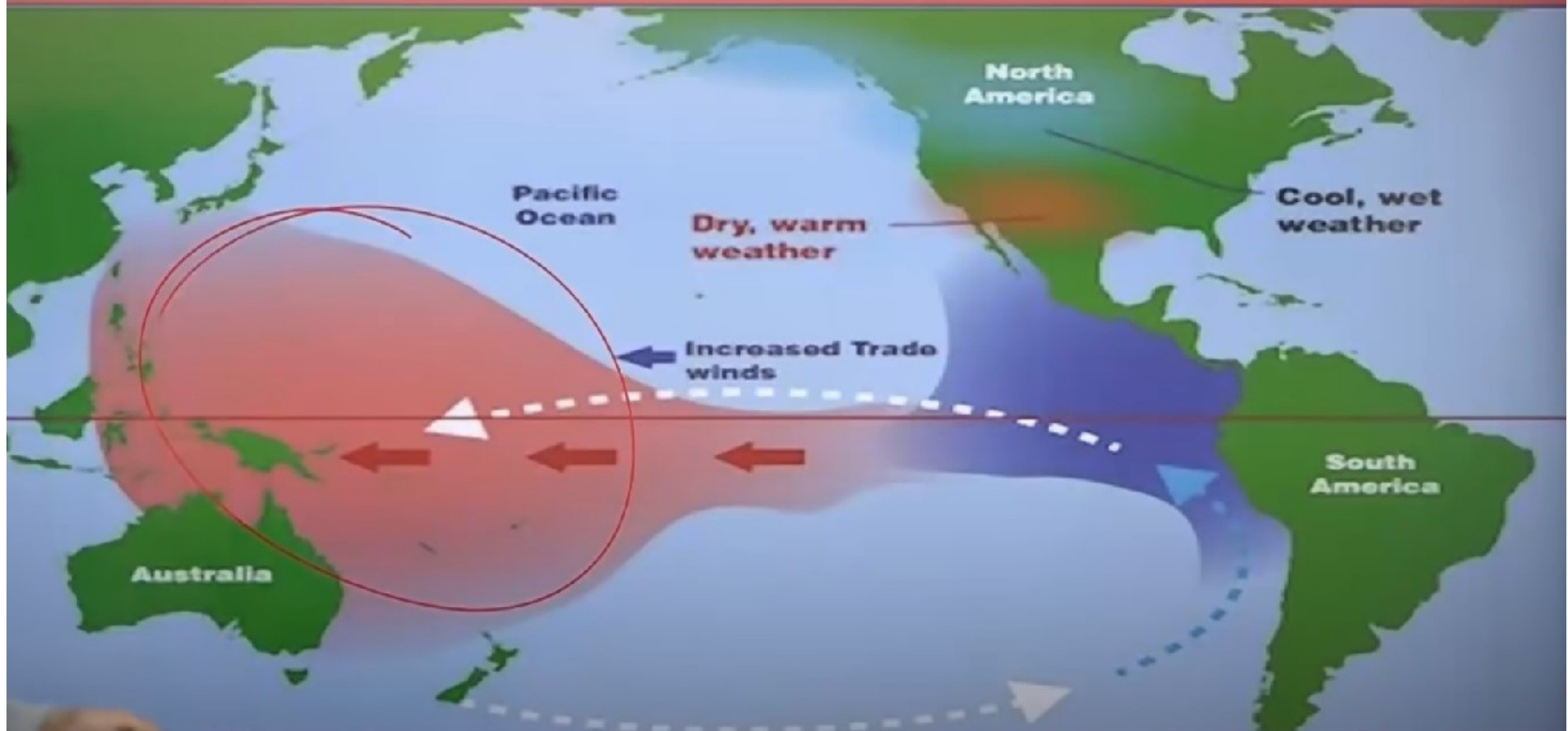
Temperature in °C

Mechanism of La Niña Process

- La Niña is sometimes referred to as 'The Little Girl', anti-El Niño, or simply 'a cold event'.
- Generally La Niña follows an El Niño. It is characterized by unusually cold ocean temperatures in the Equatorial Pacific.



Mechanism of La Niña

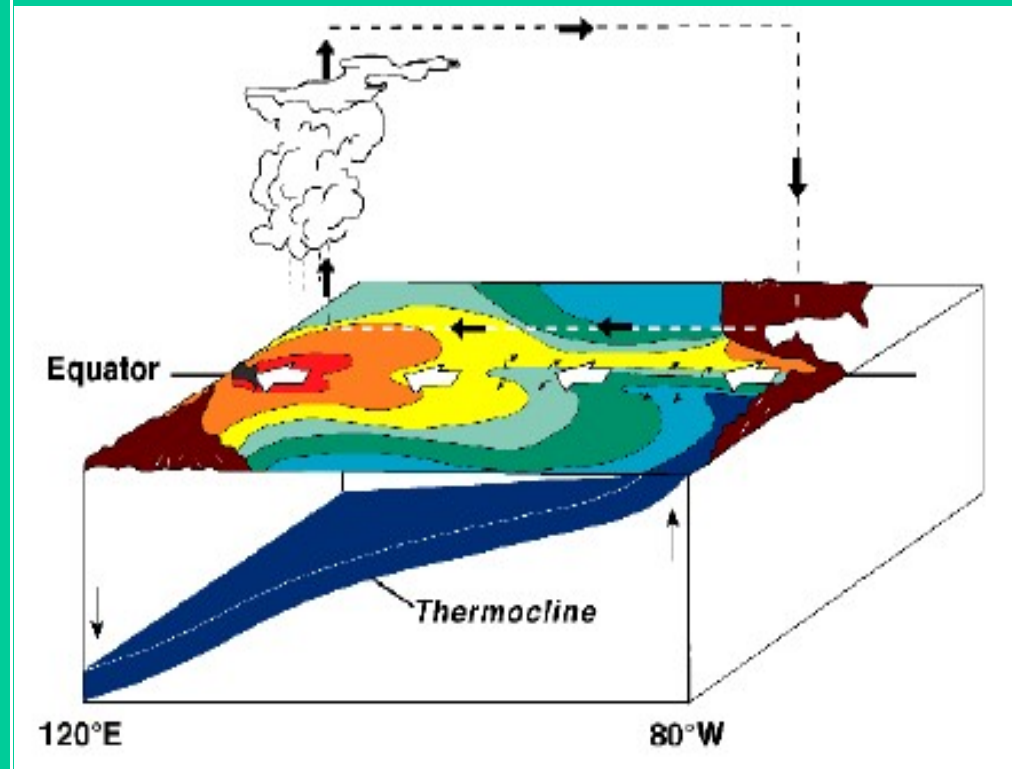


Increase in the intensity of Trade Winds. Due to this, upwelling intensifies and temperature further drops down in the eastern Pacific, whereas, increased temperature is observed in the western Pacific. Therefore, Walker Cells intensify.

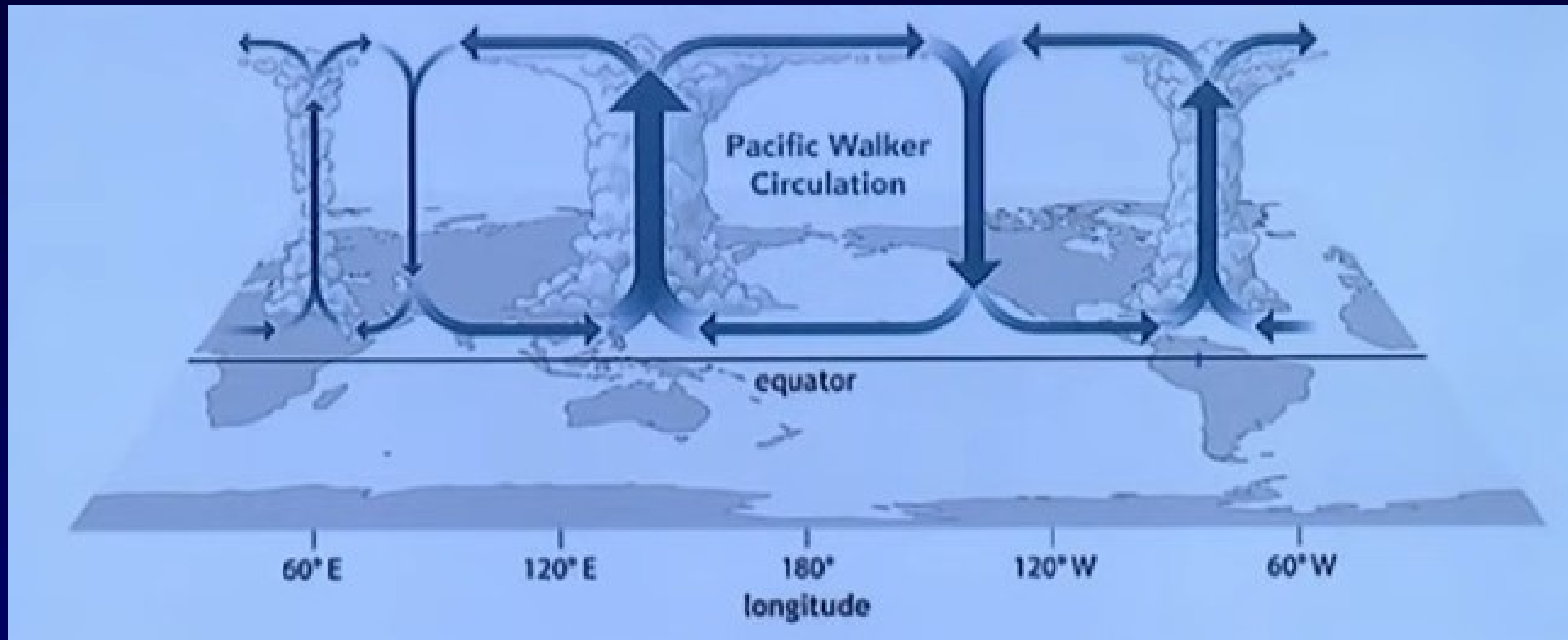
Mechanism of La Niña Process.....Contd

La Niña is a process during which the Easterly trade winds becomes stronger than normal diving more intense upwelling along the coast of Peru.

Stronger than normal South Easterly winds drive stronger offshore advection in the Eastern Equatorial Pacific along the coast of Peru which moves the thermocline much closer to the surface there by supplying more colder waters to the surface. The region of cold water along the Equator (cold tongue extends further West.



Mechanism of La Niña Process.....Contd



The Walker circulation, also known as the Walker cell, is an atmospheric system of air flow in the equatorial Pacific Ocean. This circulation, which is roughly consistent with observations, is caused by differences in heat distribution between ocean and land. It was discovered by Sir Gilbert Thomas Walker, an English Physicist and Statistician of 20th Century.

Walker Circulation

- The Walker circulation is caused by the pressure gradient force that results from a high pressure system over the eastern Pacific Ocean, and a low pressure system over Indonesia.
- The Walker circulations of the tropical Indian, Pacific, and Atlantic basins result in westerly surface winds in northern summer in the first basin and easterly winds in the second and third basins.
- As a result, the temperature structure of the three oceans display dramatic asymmetries. The equatorial Pacific and Atlantic both have cool surface temperatures in northern summer in the east, while cooler surface temperatures prevail only in the western Indian Ocean. These changes in surface temperature reflect changes in the depth of the thermocline.

- Changes in the Walker circulation with time occur in conjunction with changes in surface temperature.
- Some of these changes are forced externally and other changes appear to be the result of coupled ocean-atmosphere feedback.
- During non-El Niño conditions, the Walker circulation is seen at the surface as easterly trade winds that move water and air warmed by the sun toward the west.
- This also creates ocean upwelling off the coasts of Peru and Ecuador and brings nutrient-rich cold water to the surface, increasing fishing stocks.
- The western side of the equatorial Pacific is characterized by warm, wet, low-pressure weather as the collected moisture is dumped in the form of typhoons and thunderstorms. The ocean is some 60 cm (24 in) higher in the western Pacific as the result of this motion.

La Niñas occurred in

1903–04	1973–76
1906–07	1983–85
1909–11	1988–89
1916–18	1995–96
1924–25	1998–2001
1928–30	2005–06
1938–39	2007–08
1942–43	2008–09
1949–51	2010–12
1954–57	2016
1964–65	2017–18
1970–72	2020–22

There was a strong La Niña episode during 1988-1989.

During the 2007-2008, La Niña event was the strongest since the 1988-1989 event.

Impact of La Niña Globally



The impacts of La Niña on the global climate and ocean temperature tend to be opposite those of El Niño.

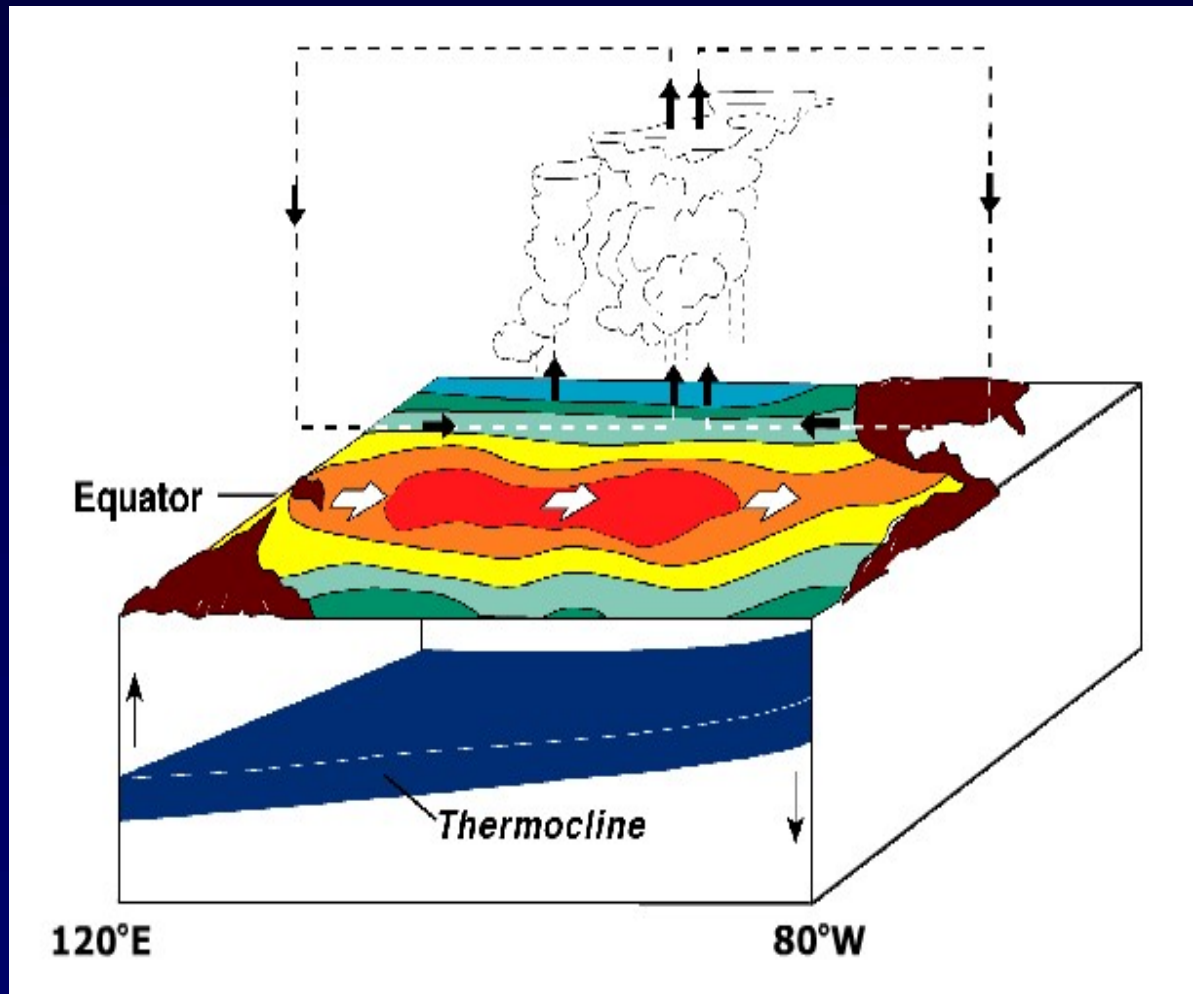
Snow and rain is experienced on the west-coast and unusually cold weather in Alaska.

During this period, there are higher than normal number of hurricanes in the Atlantic.

El Niño

El Niño is a process during which the warm waters from the Western Equatorial Pacific spread towards the Eastern Equatorial Pacific.

- This warm water depresses the thermocline in the Eastern Equatorial Pacific and thereby suppressing the upwelling.
- During El Niño the Convection Shifts from W. Equatorial Pacific towards Central Equatorial Pacific
- This condition occurs once in 4 to 7 years.



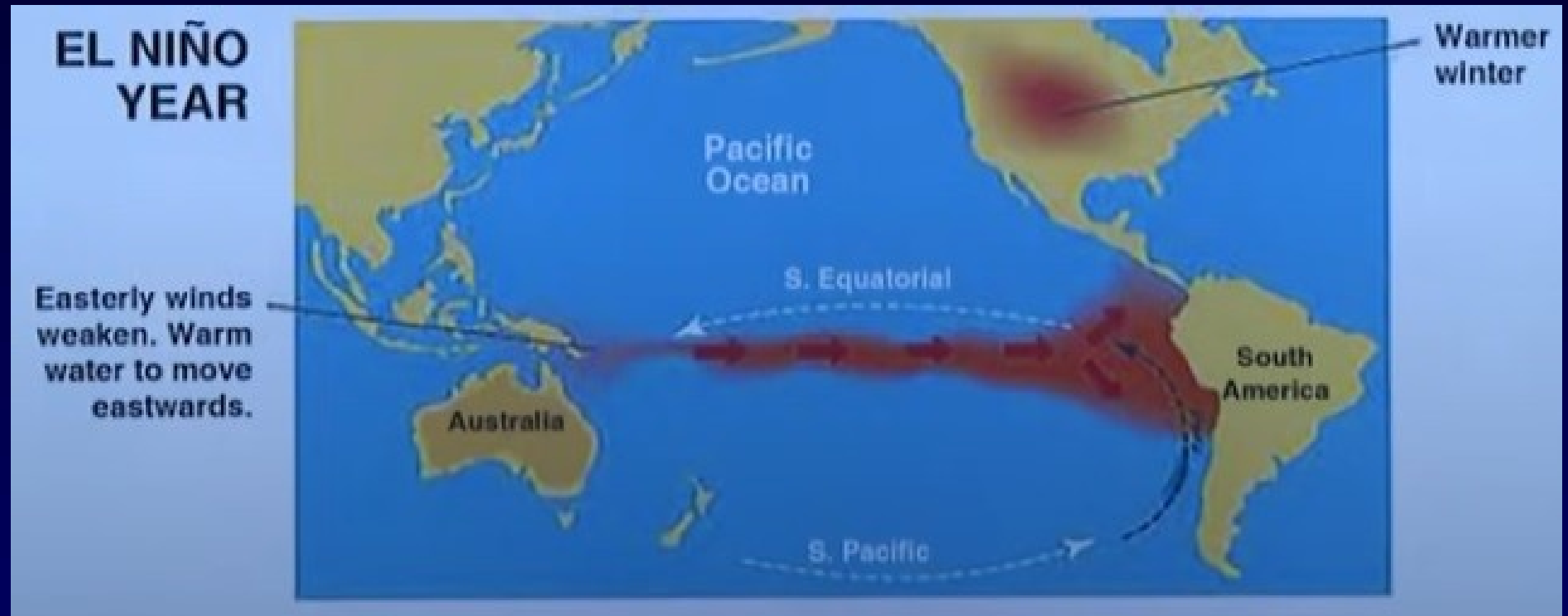
El Niño

- El Niño means 'The Little Boy', or the Christ Child in Spanish.
- El Niño was originally recognized by fishermen off the coast of South America in the 1600s, with the appearance of unusually warm water in the Pacific Ocean.
- The name was chosen based on the time of year (around December) during which these warm waters events tended to occur. The name was given by Peruvian fishermen to a period of warm waters and poor fishing that often coincided with the Christmas season.
- The term El Niño refers to the large-scale ocean-atmosphere climate interaction linked to a periodic warming in sea surface temperatures across the central and east-central Equatorial Pacific.
- The presence of El Niño can significantly influence weather patterns, ocean conditions, and marine fisheries across large portions of the globe for an extended period of time.

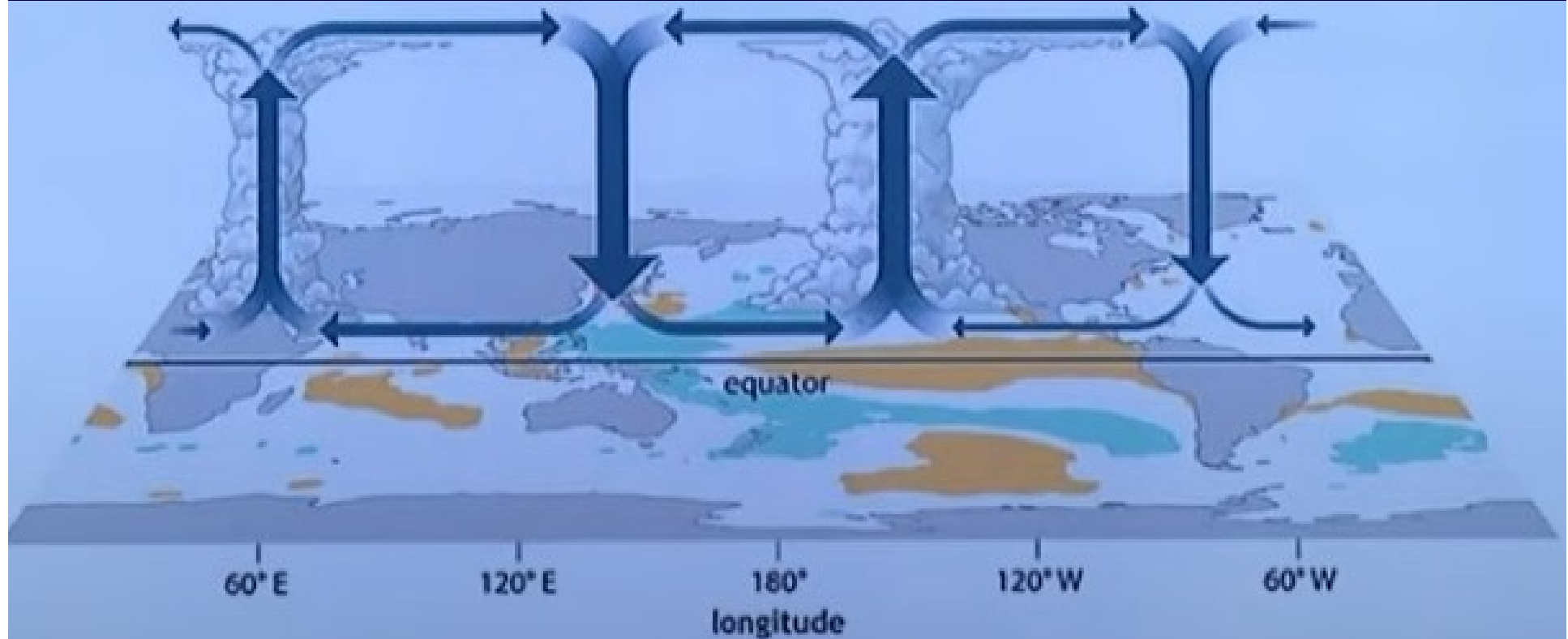
El Niño.....Contd

- When the trade winds weaken or stop, the equatorial current weakens or stops. Deep water and nutrients can no longer rise to the surface. Phytoplankton, zooplankton, and fish die or move away.
- El Niño typically persists for 12 to 18 months and recurs approximately every two to seven years.

Mechanism of El Niño Process



Mechanism of El Niño Process



Anti-Walker Current during El Nino phase in the equatorial Pacific

Recorded History of El Niño

El Niño was first recorded in the 1500's and has come again with some of its worst weather in 1997 (Another severe occurrence of El Niño was in 1982 to 1983, the worst of the 20th century).

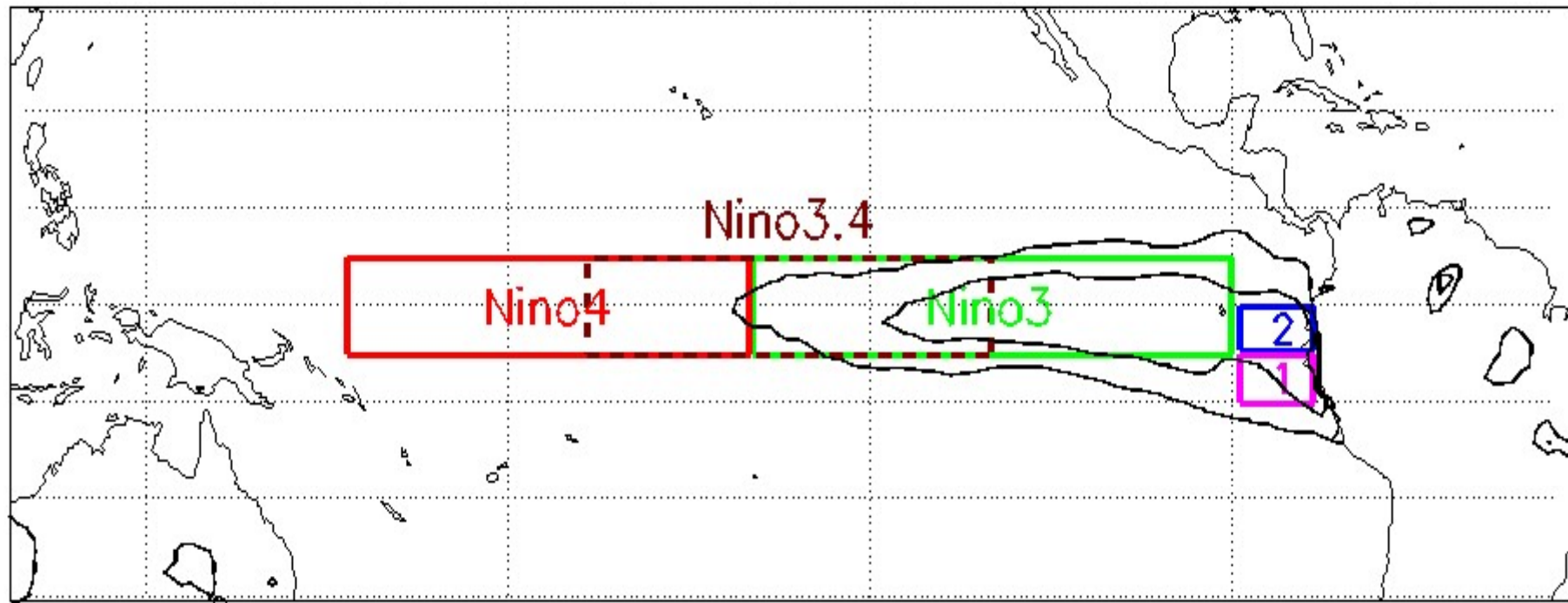
1902-1903	1953-1954	1986-1987
1905-1906	1957-1958	1991-1992
1911-1912	1965-1966	1994-1995
1914-1915	1969-1970	1997-1998
1918-1919	1972-1973	2002-2003
1923-1924	1976-1977	2004-2005
1925-1926	1982-1983	2006-2007
1930-1931	1986-1987	2009-2010
1932-1933	1991-1992	2014-2016
1939-1940	1994-1995	2018-2019
1941-1942	1997-1998	
1951-1952		

Major ENSO events were recorded in the years 1790–93, 1828, 1876–78, 1891, 1925–26, 1972–73, 1982–83, 1997–98, and 2014–16.

El Niño "Modoki"

- El Niño "Modoki" occurs in the Central-Pacific. Modoki is Japanese name means "similar, but different".
- The traditional El Niño, also called Eastern Pacific El Niño, involves temperature anomalies in the Eastern Pacific.
- However, in the last two decades non-traditional El Niños were observed, in which the usual place of the temperature anomaly (Nino 1 and 2) is not affected, but an anomaly arises in the Central Pacific (Nino 3, 4).
- The phenomenon is called Central Pacific El Niño "dateline" El Niño (because the anomaly arises near the dateline), or El Niño "Modoki".

El Niño "Modoki"



Map showing Niño 3, 4 and other index regions

The recent discovery of El Niño Modoki is being linked by some scientists to global warming. Since the Satellite data only goes back to 1979, therefore it necessitates more research must be done to find the correlation and study past El Niño episodes. The first recorded El Niño that originated in the Central Pacific and moved towards the East was in 1986.

El Niño "Modoki" events occurred in:

1991-92

1994-95

2002-03

2004-05

2009-10

The strongest such Central Pacific El Niño occurred in 2009-2010

Impacts of El Niño

1. Flood
2. Draughts
3. Coral Bleaching
4. Poor Agriculture
5. Food Inflation
6. MSP Issue (Budget Deficit)
7. Poverty (BPL)
8. Livelihood (Fishing Industry)

Impact of El Niño

Normally, trade winds blow towards the West, across the Pacific, pushing warm surface water away from the South American coast towards Australia and The Philippines.

Along the Peruvian coast the water is cold and nutrient-rich, supporting high levels of primary productivity, diverse marine ecosystems, and major fisheries.

During El Niño, the trade winds relax in the equatorial Pacific. This causes the warm water to advect from the West to East. This suppresses upwelling along the Peru coast and significantly reduces the nutrient supply to the upper ocean, thus leading to the killing of plankton and other aquatic life such as fish and cause starvation of many seabirds.

Impact of El Niño on Peruvian Economy

During La Nina

Along the West coast of South America (Peru coast) the upwelling of cold, nutrient-rich water sustains large fish populations (Peruvian Anchovy), which in turn support abundant sea birds (Guana). Droppings of Guana bird is used in the fertilizer industry. Thus, both the Anchovy as well as Guana birds contributes largely to the Peruvian economy.



<http://www.google.co.in/search?peruvian+guanna>



Anchovy

Engraulis ringens

<http://www.google.co.in/search?peruvian+anchovy>

El Niño reduces the upwelling along the Peru coast leading to killing of plankton and fish (Peruvian Anchovy) and the starvation of sea birds (Guana). Both of the above adversely affects the Peruvian Economy.

Impact of El Niño Globally

El Niño causes a wide variety of disasters globally. For example, famine in Indonesia in 1983, bush fires in Australia arising from droughts, rainstorms in California, and the destruction of anchovy fishery off the coast of Peru.



<http://www.google.co.in/searchforest+fire>

Bush fire in Australia



<http://www.windows2universe.org>

This area of Lakeport, California was flooded due to extreme weather during the 1998 El Niño event.

The impact of the 1997/98 spell of the phenomena was very damaging

- Floods devastated the America
- Storms hit China
- Drought parched Austria
- Forest fires burnt parts of South-East Asia and Brazil
- Indonesia experienced the worst drought in the last 50 years
- In Mexico, the town of Guadalajara saw snow for the first time since 1881
- In the Indian Ocean, it affected the movement of the Monsoon winds



<http://standeyo.com/NEWS>

A parched paddy field, blamed on El Niño, in Malaysia.

Southern Oscillation

The Southern Oscillation describes a bimodal variation in sea level barometric pressure between observation stations at Darwin, Australia and Tahiti. It is quantified in the Southern Oscillation Index (SOI), which is a standardized difference between the two barometric pressures.

The Southern Oscillation is the accompanying atmospheric component, coupled with the sea temperature change: El Niño is accompanied by high air surface pressure in the tropical western Pacific and La Niña with low air surface pressure there. The two periods last several months each and typically occur every few years with varying intensity per period.

Southern Oscillation Index (SOI)

- The SOI tracks the atmospheric part of the El Niño - La Niña climate pattern by comparing surface air pressure anomalies at Darwin, Australia, to pressure anomalies at Tahiti, French Polynesia.
- The anomalies - departures from average conditions - indicate the strength of the Walker Circulation.
- A positive SOI indicates La Niña, while a negative SOI indicates El Niño.

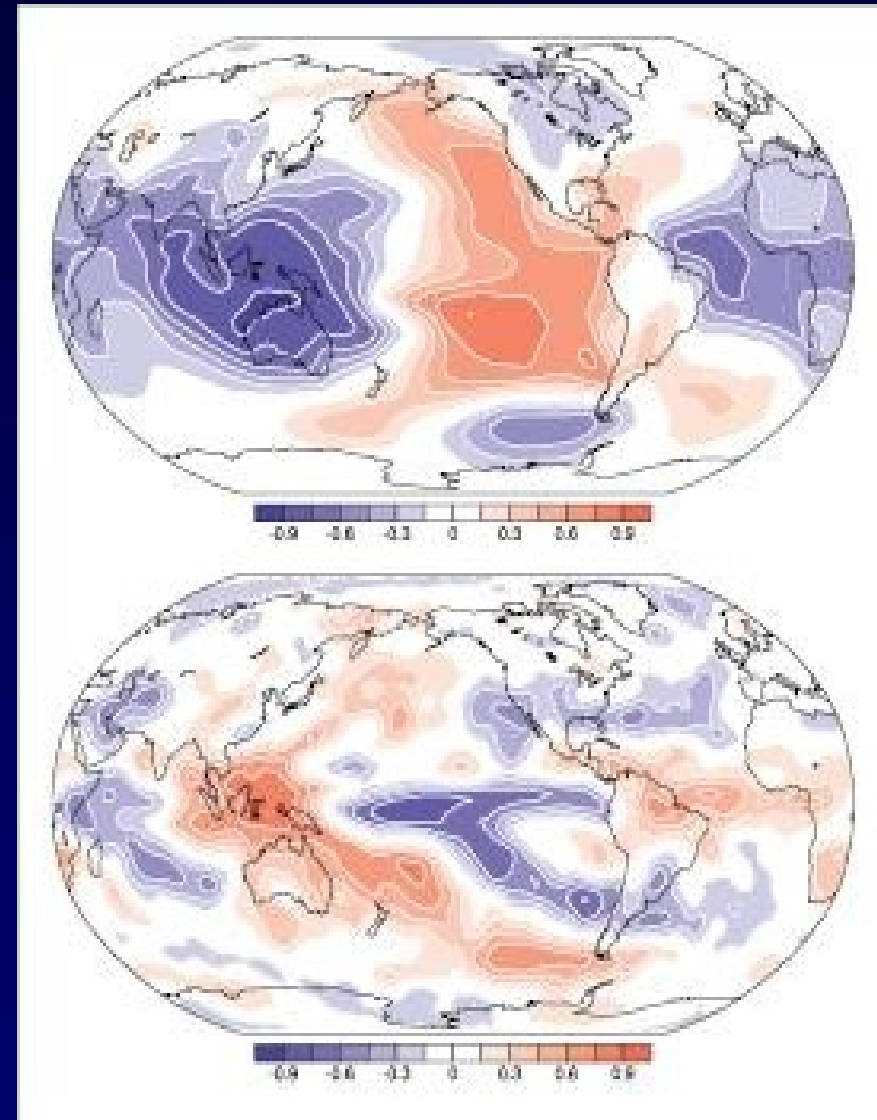
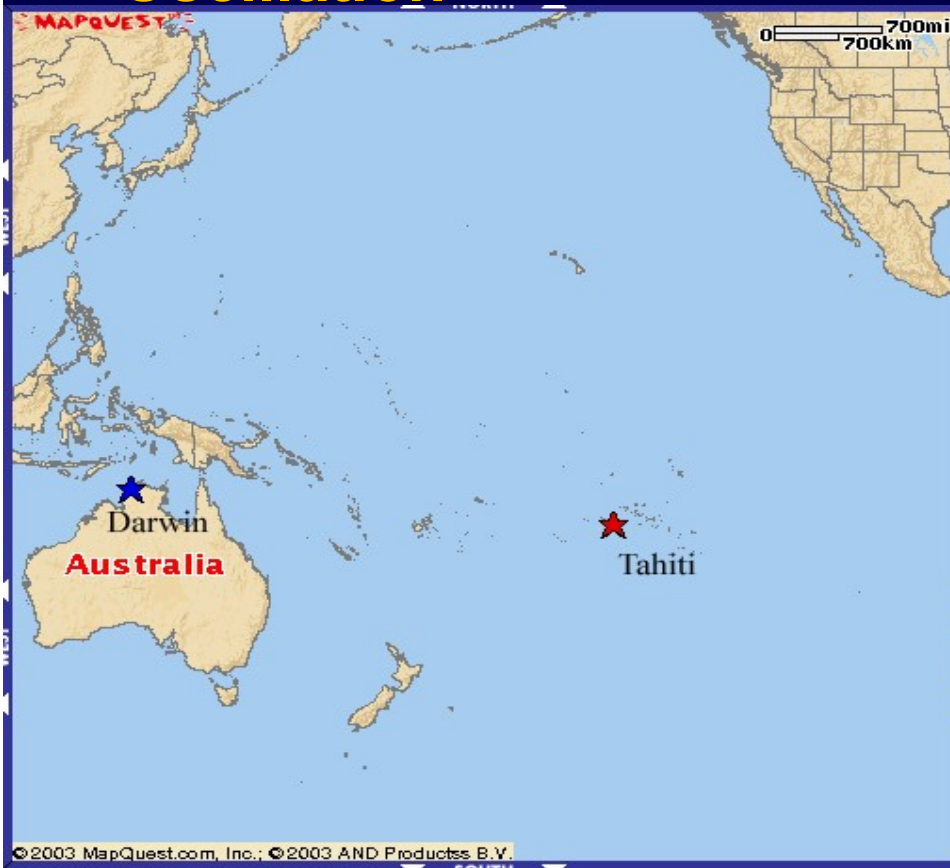
Southern Oscillation Index (SOI)

Positive SOI	Negative SO
Pressure at Tahiti is grater than that at Darwin	Pressure at Darwin is grater than that at Tahiti
High pressure over eastern Pacific	Low pressure over eastern Pacific
Drought condition in the Eastern Pacific and good rainfall in western Pacific	Drought condition in the western Pacific and good rainfall in eastern Pacific
Good for Indian Monsoon	Bad for Indian Monsoon

Southern Oscillation.....Contd

1. Rise in surface air pressure over Darwin (Australia)
2. Fall in air pressure over Tahiti

This is known as Southern Oscillation



ENSO

El Niño - Southern Oscillation (ENSO) is an irregular periodic variation in winds and sea surface temperatures over the tropical eastern Pacific Ocean, affecting the climate of much of the tropics and subtropics. The warming phase of the sea temperature is known as El Niño and the cooling phase as La Niña.

The ENSO is a single climate phenomenon that periodically fluctuates between three phases: Neutral, La Niña and El Niño. La Niña and El Niño are opposite phases which require certain changes to take place in both the ocean and the atmosphere before an event is declared.

Mechanisms that cause the oscillation remain under study. The extremes of this climate pattern's oscillations cause extreme weather (such as floods and droughts) in many regions of the world. Developing countries dependent upon agriculture and fishing, particularly those bordering the Pacific Ocean, are the most affected.



Pillow lava at CIR observed during manned submersible dive